REALTY360 AUSTIN

Application Architecture Report

System design, data flows, and engineering decisions for Travis County property analysis

Version 1.4 July 13, 2026 | Austin / Travis County, TX

MLS SQLite | TCAD Neon Cache | Unified Comps | Flip Engine | Policy B Tax

Technology stack

- Client: Vite + React + shadcn/ui
- API: Express (TypeScript)
- Domain: shared/ - comps, flip, TCAD, property-profile
- Data: data/mls.sqlite + Neon tcad_parcels

Contents

Diagrams are embedded inline throughout the report. The markdown appendix uses ASCII placeholders; this PDF renders vector architecture diagrams instead.

> Realty360 Austin - Application Architecture Report - Version 1.4 for Austin / Travis County. See following sections for system design and data flows.

1. Executive Summary

Realty360 Austin is a full-stack property analysis tool for Travis County real estate investors and analysts. It combines:

- MLS closed and open listings (sale comparables, subject lookup)
- TCAD tax parcel data (assessed/appraised values, parcel centroids)
- Unified comparables search (closed sales, active listings, optional tax-reference parcels; MLS comps enriched with TCAD tax fields)
- Property profile composition with in-process cache and multi-parcel TCAD (taxCandidates)
- MLS <-> TCAD crosswalk (precomputed listing_key <-> prop_id in Neon)
- Post-flip deal analysis (ARV from comps, rehab tiers, hold/financing costs with Policy B multi-parcel tax, margin viability)

The app is built as Vite + React (client) and Express (API), with shared TypeScript logic in shared/. Production serves a single Node process; development runs Vite on port 3000 proxying /api to Express on 3001.

What it is not (yet): a production MLS Grid proxy for the browser, a machine-learning ARV model, or full ACTRIS MLS history in the local seed.

2. System Architecture

2.1 High-Level Topology

SYSTEM TOPOLOGY

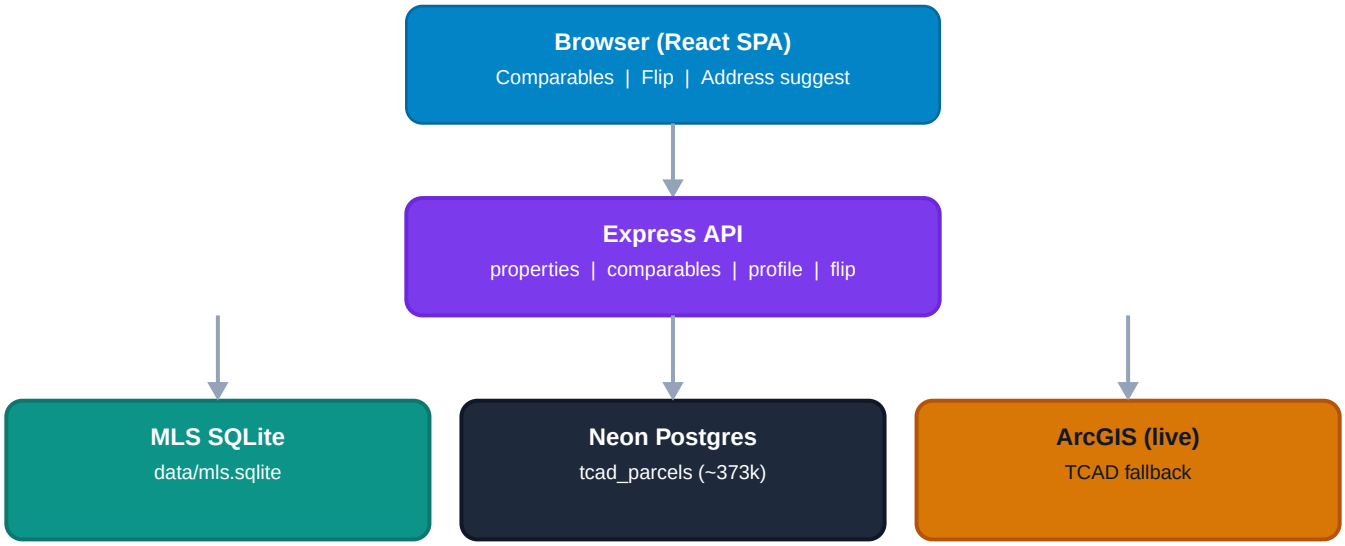

Figure 1 - Browser, API, and data store layers

2.2 Request Path (Development)

DEVELOPMENT REQUEST PATH

Figure 2 - Development ports and proxy path

Component	Port	Role
Vite dev server	3000	UI, proxies /api -> 3001
Express dev:api	3001	JSON API, reads SQLite + Postgres
pnpm start (prod)	3000	Bundled static + API (or API-only if no build)

2.3 Shared Logic

Business rules live in shared/ so the same comp-matching, flip math, TCAD address scoring, and DTO composition run in scripts, API routes, and (where applicable) the client via @shared alias.

3. Repository Layout

Path	Purpose
client/	React UI (Comparables landing, flip results, shadcn components)
server/	Express app, route handlers
shared/	Domain logic: MLS, TCAD, comparables, property-profile, flip
shared/env/load-env-local.ts	Loads .env.local for Node scripts and Express
shared/mls/sqlite.ts	MLS SQLite schema and query helpers (runtime)
scripts/	Seed jobs (seed-mls, seed-tcad, seed-crosswalk), smoke-flip, architecture PDF
data/	Gitignored runtime data (mls.sqlite, raw JSON dumps)
docs/	Roadmap and architecture documentation

4. Data Sources & Storage

4.1 MLS (Multiple Listing Service)

Aspect	Detail
Upstream	MLS Grid API (https://api-demo.mlsgrid.com/v2 or production URL)
Originating system	actris (Austin Board of Realtors) - shared/mls/constants.ts
Local store	SQLite data/mls.sqlite, table properties
Seed command	pnpm seed:mls (closed + active/pending residential)
Runtime	API reads SQLite only; no live MLS Grid on user requests
Token	MLS_GRID_TOKEN or VITE_MLS_GRID_TOKEN in .env.local (seed only)

Row shape: normalized address, lat/lon, beds/baths/sqft, list/close price, close date, pool, garage, lot acres, property condition, status.

4.2 TCAD (Travis Central Appraisal District)

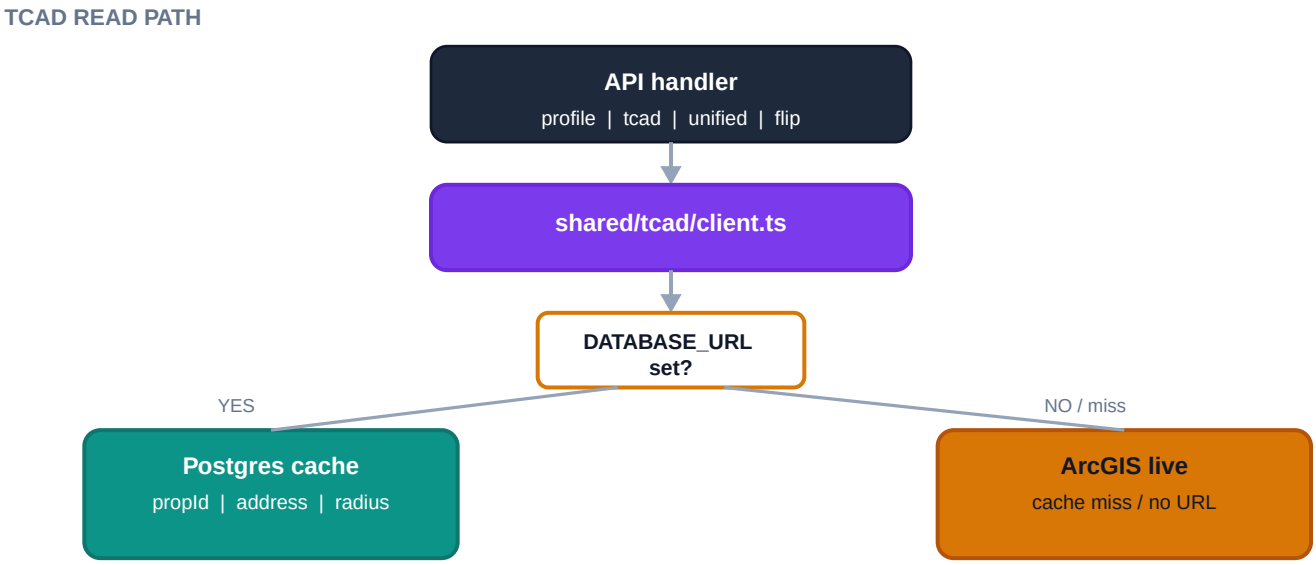

Figure 7 - Cache-first TCAD read path

Aspect	Detail
Upstream	ArcGIS MapServer layer TCAD/MapServer/0 (full county parcels)
Local store	Postgres tcad_parcels (Neon) via DATABASE_URL

Seed command	pnpm seed:tcad (~387k parcels, paginated ETL)
Runtime	Cache-first in shared/tcad/client.ts; live ArcGIS fallback
Coverage	Travis County only - Williamson/Georgetown addresses return not_found

Fields cached: PROP_ID, geo_id, situs address, tax values (appraised, market, assessed), acres, WGS-84 centroid.

4.3 MLS <-> TCAD Crosswalk

Aspect	Detail
Store	Neon Postgres table mls_tcad_crosswalk (same DATABASE_URL as TCAD)
Seed command	pnpm seed:crosswalk (after seed:mls + seed:tcad); --fresh truncates
Row shape	listing_key, prop_id, address_norm, match_method, match_score
Build logic	Cache-only TCAD address lookup per MLS listing; single + ambiguous outcomes stored
Runtime	Comp enrichment and property profile resolve TCAD via crosswalk before address scoring

Current seed stats (6,143 listings): 1,757 distinct listings linked (~29%); 1,919 total rows (includes multi-parcel ambiguous).

4.4 Data Independence Principle

MLS and TCAD remain separate upstream sources. The crosswalk is a precomputed optimization layer - not all MLS addresses link to a TCAD situs (~71% still not_found after address matching).

5. Core User Flows

5.1 Comparables Search

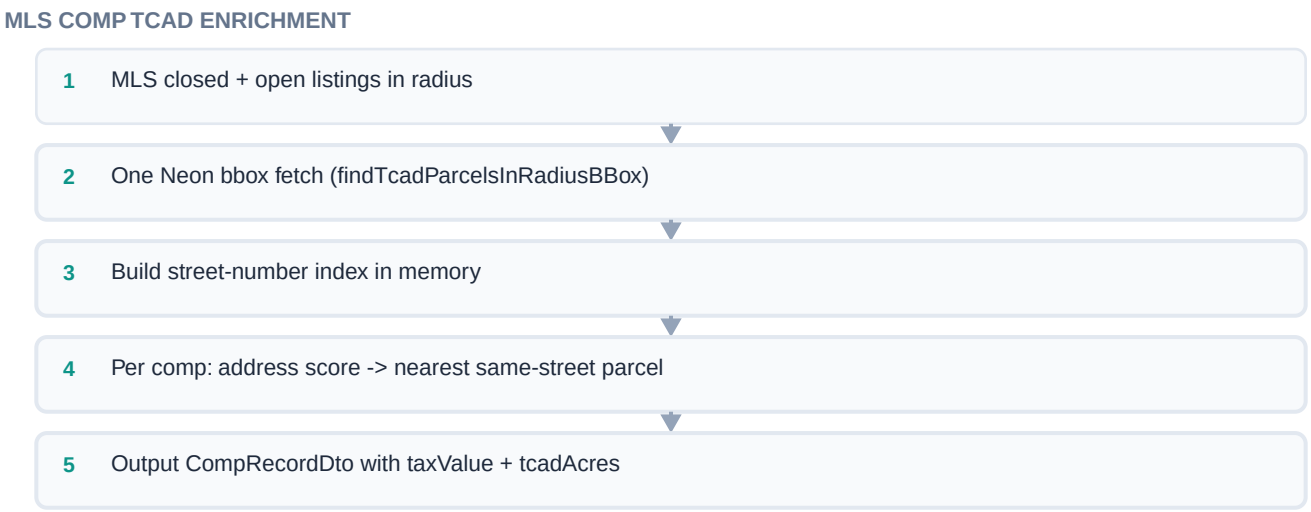

Figure 4 - Batch TCAD enrichment for MLS comps

1. User enters address (with typeahead from MLS SQLite).
2. UI calls GET /api/comparables/unified with radius, recency, optional includeTcad, optional propId.
3. Server loads subject via fetchPropertyProfileCached (MLS + TCAD profile-first); resolves MLS row for comp filters.
4. Returns sections:

- Closed sales (MLS) - sale comps, each enriched with TCAD taxValue and tcadAcres when a parcel matches - Active listings (MLS open status) - same TCAD enrichment on listing comps - Tax references (TCAD radius, includeTcad=true) - reference only, not sale comps

1. Response includes subjectProfile for subject card tax display and flip cache reuse.

Subject card: shows TCAD tax value, acres, deed date when single match; lists all taxCandidates with combined tax when ambiguous; optional parcel picker re-runs search with propId.

TCAD enrichment (closed + open comps): batch crosswalk lookup by listing_key (single link only), then strict situs address match against one radius prefetch (findTcadParcelsInRadiusBBox). Tax fields attach only when a real TCAD parcel matches - no nearest-parcel coordinate fallback. UI citation: MLS + TCAD when enriched.

Comps are bucketed into exclusive mile rings (0-0.5, 0.5-1, 1-2 mi, ...) with match % from beds/baths/distance.

5.2 Property Profile

PROPERTY PROFILE COMPOSITION

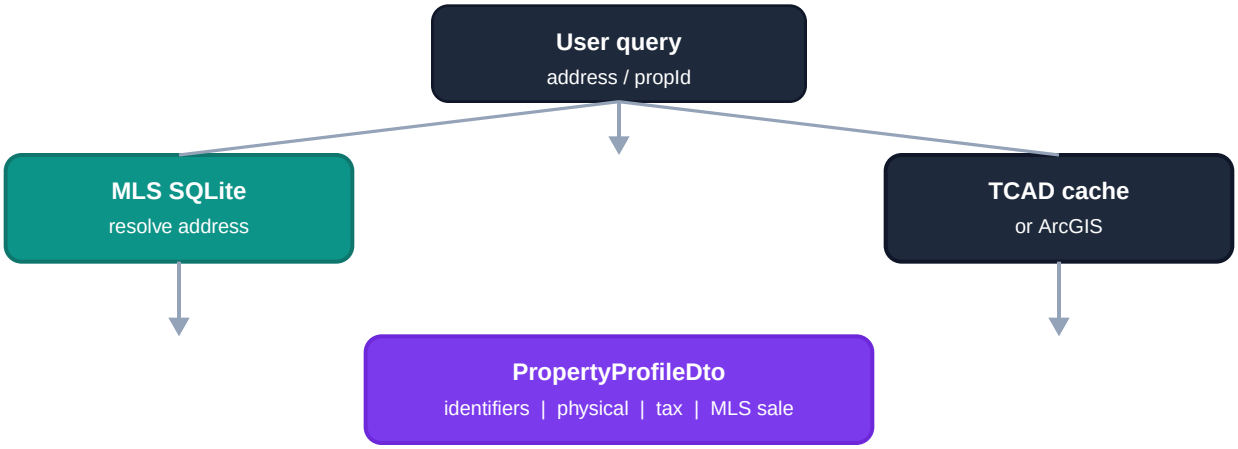

Figure 5 - MLS + TCAD profile composition

MULTI-PARCEL TCAD HANDLING

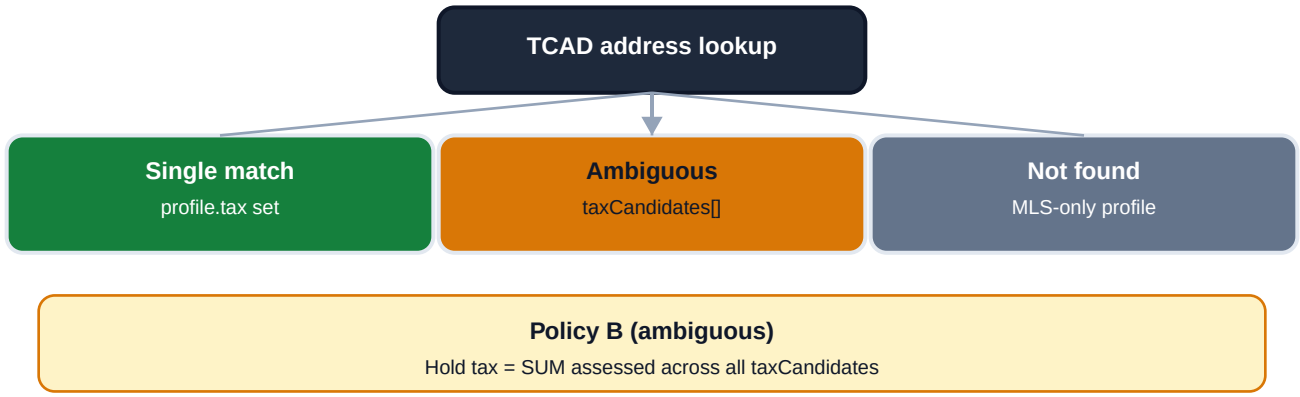

Figure 6 - Multi-parcel TCAD outcomes and Policy B

1. GET /api/property/profile?address= or ?propId= (or both)
2. fetchPropertyProfile() / fetchPropertyProfileCached() loads MLS by address.
3. TCAD linking priority: user propId -> MLS listing_key crosswalk -> address lookup (cache -> ArcGIS).
4. composePropertyProfile() merges identifiers, physical, MLS sale, tax, location.
5. Single TCAD match -> tax set, tcadMatch.status: "single".
6. Ambiguous TCAD match -> tax: null, taxCandidates[], tcadMatch.status: "ambiguous" (e.g. duplex - multiple prop_id at same situs).
7. No TCAD match -> MLS-only profile, tcadMatch.status: "not_found"; comparables and flip continue without tax fields.
8. Address matching tolerates MLS-style queries (504 Bramble Dr, Austin, TX 78745) by excluding city/state tokens from situs scoring (shared/tcad/address-query.ts).

Profile cache: in-process Map, 30 min TTL, keys address:{norm} and/or propId:{id}|address:{norm}. Comparables search warms cache; flip reuses same profile without re-fetching TCAD.

5.3 Flip Analysis

1. User runs comparables search, switches to Flip analysis tab.

2. Enters purchase price and rehab scope tier (cosmetic / moderate / full).
3. POST /api/predict/flip in enriched mode:

- Loads property profile via fetchPropertyProfileCached (address only - preserves taxCandidates for Policy B even if UI selected a propId for display) - Derives ARV from similar closed MLS comps (median $/sqft x subject sqft) - Computes rehab, closing, hold, financing - Hold tax basis (Policy B): single parcel -> assessed -> appraised; ambiguous -> sum assessed across all taxCandidates; no TCAD -> purchase price - Returns viability band: strong / marginal / weak / negative

Manual mode: user supplies explicit arv + livingAreaSqft without subject lookup.

6. API Surface

Method	Endpoint	Purpose
GET	/api/health	Liveness check
GET	/api/properties/suggest	Address typeahead (MLS)
GET	/api/properties/by-address	Single MLS property
GET	/api/properties/by-radius	Deprecated - closed comps only; use /api/comparables/unified
GET	/api/comparables/unified	Closed + open + optional TCAD; optional propId; returns subjectProfile
GET	/api/property/profile	Composed MLS + TCAD profile
GET	/api/tcad/property	TCAD parcel by propId or address
POST	/api/predict/flip	Rules-based flip prediction

All routes are mounted in server/app.ts.

> Deprecation: GET /api/properties/by-radius responds with Deprecation: true, a Warning header, and a deprecation object in the JSON body. Prefer GET /api/comparables/unified for all new integrations.

7. Domain Modules

7.1 shared/mls/

- RESO property types, MLS Grid client, address normalization
- shared/mls/sqlite.ts: SQLite schema, radius queries, address resolution

7.2 shared/tcad/

- ArcGIS client with retry/timeout
- Address fuzzy matching (address-query.ts)
- Postgres cache (db.ts)
- MLS <-> TCAD crosswalk (crosswalk.ts)
- Parcel centroids from polygon geometry

7.3 shared/comparables/

- match-score.ts - mile rings, match %, distance score
- comp-record.ts - unified CompRecordDto with source, compRole, taxValue, tcadAcres
- tcad-enrichment.ts - crosswalk batch lookup + strict situs address match for MLS sale/listing comps
- unified-search.ts - orchestrates MLS + TCAD comp sections and enrichment

7.4 shared/property-profile/

- fetch-profile.ts - enrichment orchestration (MLS + crosswalk + TCAD, ambiguous handling)
- compose.ts - merges partial MLS/TCAD into PropertyProfileDto
- profile-cache.ts - in-process TTL cache keyed by address and/or propId
- tcad-tax.ts - pickProfileTaxValue, taxHoldBasisFromProfile (Policy B sum for multi-parcel)
- Provenance and completeness scoring

7.5 shared/flip/

- config.ts - Travis County rehab tiers, closing %, hold/financing defaults
- arv-comps.ts - similarity filter, tier-based $/sqft slice (top half / quartile)
- deal-costs.ts - deterministic cost stack
- predict-flip.ts - end-to-end prediction pipeline

8. Flip Prediction Engine

8.1 ARV (After Repair Value)

Source: MLS closed sales within radius, filtered by:

- Recency (maxAgeMonths on close_date)
- Similarity: +/-25% sqft, beds/baths within delta rules
- Rehab tier slice: cosmetic uses all similar; moderate uses top 50% $/sqft; full uses top 25%

```
ARV = median(pricePerSqft of slice) x subjectLivingAreaSqft
```

Override: request body arv -> manual mode.

8.2 Cost Stack

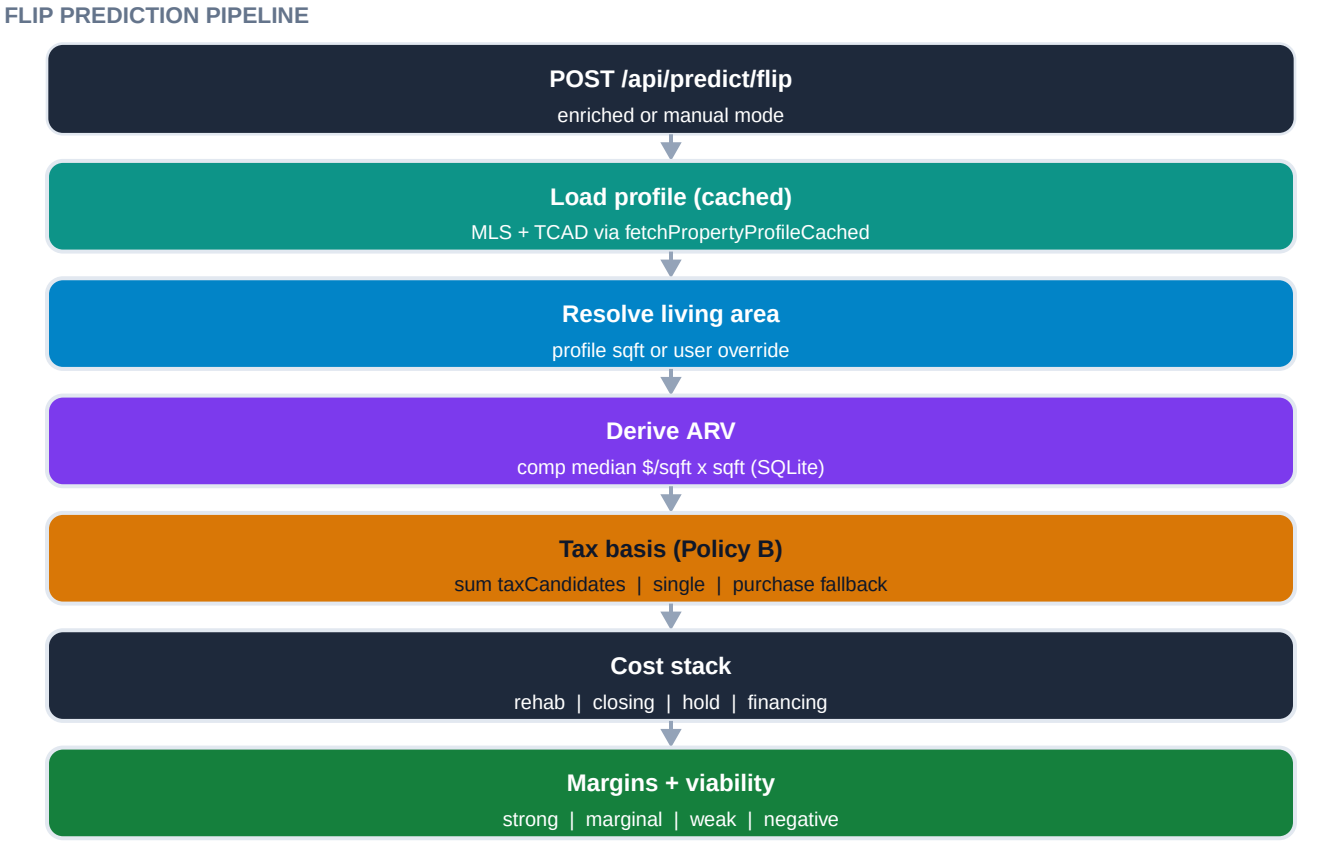

Figure 8 - Flip prediction pipeline

Component	Basis
Rehab	tier $/sqft x living area (sqft from MLS profile)
Buy closing	2.5% of purchase price (user input)

Hold - property tax	Policy B: sum assessed across taxCandidates when ambiguous; else single assessed -> appraised -> purchase price; x 2.2% annual x hold months
Hold - insurance / utilities / HOA	fixed monthly amounts from deal config
Financing	hard-money style: LTC ratio, points, interest during hold
Sell closing	7% of ARV (ARV from MLS comps)

8.3 Viability Bands

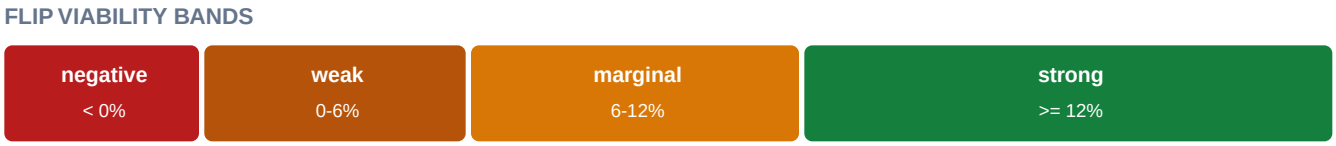

Figure 9 - Net margin viability bands

Net margin % (of ARV)	Label
>= 12%	strong
6-12%	marginal
0-6%	weak
< 0%	negative

8.4 Explicit Non-Goals

- TCAD appraised value is not used as ARV or sale comp
- Tax values affect hold cost only (when TCAD resolves)

9. Key Design Decisions

Decision	Rationale
MLS in SQLite, not live API	Fast radius queries; avoids MLS Grid rate limits on every request
TCAD in Postgres (Neon)	~373k parcels; eliminates per-request ArcGIS latency; one bbox query per comp search
Cache-first TCAD with ArcGIS fallback	Works before seed completes; resilient to cache gaps; `source: tcad-cache \
MLS comp TCAD enrichment	Crosswalk first (fast); strict situs match fallback; tax only when TCAD parcel exists
Profile cache (30 min TTL)	Comparables warms profile; flip reuses without duplicate TCAD/MLS fetches
Multi-parcel Policy B	Duplex/multi-prop_id situs: hold tax sums all candidate assessed values; UI lists each parcel
MLS-only fallback	TCAD not_found -> valid MLS profile; hold tax uses purchase price
MLS <-> TCAD crosswalk	Precomputed listing_key <-> prop_id in Neon; ~29% of seeded listings linked
No nearest-parcel comp tax	Avoids false tax on MLS marketing addresses (e.g. ROW parcels, Tournament/ Tourney mismatches)
MLS-style address scoring	City/state stripped from TCAD situs match so profile + flip resolve full formatted addresses
Batch comp enrichment	One radius prefetch + crosswalk prop_id batch fetch (not N fuzzy matches per comp)
TCAD tax comps are reference only	Appraised "` market sale price; prevents misleading ARV
Rules engine before ML	Auditable math; same API contract for future Python model
Shared TypeScript domain layer	One source of truth for comps, flip, and profile logic
Enriched vs manual flip modes	Supports full subject lookup or spreadsheet-style what-if
Dedupe PROP_ID on TCAD seed	ArcGIS pages can repeat prop_ids (multipart parcels)

tcad-arcgis`

10. Operational Commands

```
pnpm install
pnpm seed:mls              # MLS closed + active/pending -> data/mls.sqlite
pnpm seed:tcad             # Full county -> Neon tcad_parcels
pnpm seed:crosswalk        # MLS listing_key <-> TCAD prop_id -> Neon mls_tcad_crosswalk
pnpm dev:api               # API :3001
pnpm dev                   # UI :3000
pnpm smoke:flip            # API/engine regression checks
pnpm docs:architecture-pdf # Regenerate docs/ARCHITECTURE_REPORT.pdf
pnpm build && pnpm start   # Production-style single server
```

Environment (.env.local):

- VITE_MLS_GRID_TOKEN / MLS_GRID_TOKEN - MLS Grid bearer token
- DATABASE_URL - Neon Postgres for TCAD cache (tcad_parcels). Must match the Neon project/ branch you intend (verify ep-... hostname in connection string vs dashboard).

11. Roadmap & Gaps

Completed (Phases 1-5.3)

- MLS schema extensions, mile-bucket comps, unified comparables API
- Property profile composition, TCAD propId/address APIs, profile cache, multi-parcel taxCandidates
- Subject card TCAD tax display; optional propId parcel picker
- Flip rules engine v0, flip UI on comparables landing (FlipPredictionResults)
- Policy B hold tax (taxHoldBasisFromProfile - sum assessed for ambiguous parcels)
- TCAD Neon ETL (pnpm seed:tcad, ~373k parcels), cache-first reads (shared/tcad/db.ts)
- MLS <-> TCAD crosswalk (pnpm seed:crosswalk, mls_tcad_crosswalk, runtime in crosswalk.ts)
- MLS sale/listing comp enrichment - crosswalk + strict address match (tcad-enrichment.ts)
- MLS-style TCAD address matching (city/state excluded from scoring)
- Flip smoke tests (pnpm smoke:flip)
- Legacy GET /api/properties/by-radius deprecated (use /api/comparables/unified)
- Project cleanup: runtime code in shared/, pruned unused UI components and dead client routes

MLS <-> TCAD coverage (July 2026)

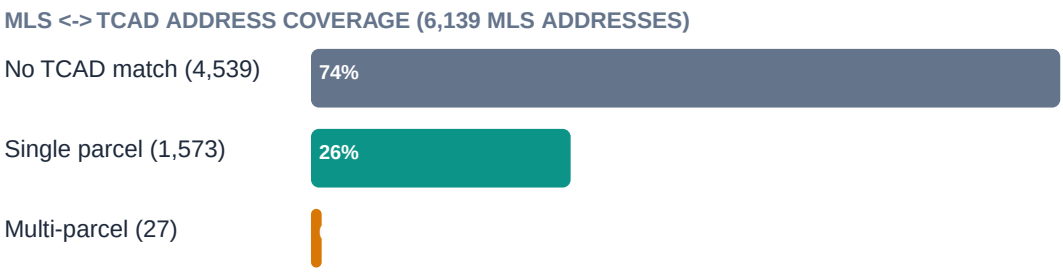

Figure 10 - MLS address crosswalk outcomes

Address lookup (6,139 distinct MLS addresses):

Outcome	Count
Single TCAD parcel	1,573
No TCAD match (MLS-only)	4,539
Ambiguous (multi-parcel)	27

Crosswalk table (6,143 MLS listings seeded):

Outcome	Count
Single link	1,716
Ambiguous listings	41

Not found	4,386
Distinct listings linked	1,757 (~29%)

Example multi-parcel: 1613 W Braker Ln #B. MLS-only examples: 507 Hammack Dr, 34 Tournament Way, 50 Tournament Way #I-52.

Planned

Item	Description
Deal ledger	Labeled flips for ML training
Python ML ARV model	Same POST /api/predict/flip contract
geoId lookup	Alternate TCAD key
Coordinate-based crosswalk seed	Link MLS lat/lon to TCAD parcel when situs address mismatches
Monthly TCAD + crosswalk refresh	Cron re-run seed:tcad then seed:crosswalk

Known Limitations

- MLS seed cap (~2k-5k rows configurable) - not full ACTRIS history
- Non-Travis addresses have MLS data but no TCAD tax
- ~71% of MLS listings have no TCAD crosswalk link (situs vs marketing address gaps, units, numbering)
- Multi-parcel ambiguous is rare (~41 listings in crosswalk seed)
- Condos/townhomes often lack unit-level TCAD situs - no tax on comp unless situs matches
- Flip ARV requires sufficient similar closed comps in radius